Review
Fuzzy Sets

Fuzzy Sets Operators

- Zadeh
  - c(a) = 1-a
  - i(a,b) = min{a,b}
  - u(a,b) = max{a,b}
- Lukasiewicz
  - c(a) = 1-a
  - i(a,b) = max{a+b-1,0}
  - u(a,b) = min{a+b,1}

Fuzzy Sets Properties

	Properties	Crisp sets	Zadeh	Lukasiewicz
(1)	Idempotence laws	Valid	Valid	Invalid
(2)	$A \cup \bar{A} = X$ and $A \cap \bar{A} = \emptyset$	Valid	Invalid	Valid
(3)	Distributivity laws	Valid	Valid	Invalid
(4)	De Morgan laws	valid	Valid	Valid

Fuzzy Sets Properties

- The idempotence properties (1) are only satisfied by Zadeh operators.

Fuzzy Sets Properties

- Any intersection and union satisfying De Morgan properties (4) are called dual with respect to the complement.

- The dual of an intersection operator is a union operator.
- The dual of a union operator is an intersection operator.

Fuzzy Sets Properties

- Any intersection and union defined on nondegenerate fuzzy sets that satisfy properties (2) of excluded middle and contradiction do not satisfy properties (3) of distributivity.
- Any intersection and union that satisfy properties (2) and (3) are defined on degenerate fuzzy sets (i.e. crisp values).

Fuzzy Partition

- If C is a fuzzy set, then $P = \{A_1, A_2, \ldots, A_n\}$ is a fuzzy partition of C if
  - $C = A_1 \cup A_2 \cup \cdots \cup A_n$
  - $A_i \cap A_j = \emptyset$ for any i and j distinct
- The following statement is true if and only if the operators are Lukasievicz:
  - P is a fuzzy partition on C if and only if

$$A_1(x) + A_2(x) + \cdots + A_n(x) = C(x), \forall x \in X$$